9. (i) Relative to a fixed origin O, the points A, B and C have position vectors **a**, **b** and **c** respectively.

Points A, B and C lie in a straight line, with B lying between A and C.

Given $AB:AC = 1:3$ show that

$$\mathbf{c} = 3\mathbf{b} - 2\mathbf{a}$$

(3)

(ii) Given that $n \in \mathbb{N}$, prove by contradiction that if n^2 is a multiple of 3 then n is a multiple of 3

(5)